Predicting Car Pricing in Germany using different Machine Learning Algorithms with AutoScout24 Dataset

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***Abstract*—This paper analyzes the application of different machine learning algorithms using a diverse set of models: Linear Regression, Ridge Regression, Random Forest, Gradient Boosting, and XGBoost with the AutoScout24 dataset from Kaggle. We conducted vigorous Exploratory Data Analysis (EDA) to understand the fundamental patterns in features such as mileage, horsepower, fuel type, and year. Understanding these patterns is crucial in enhancing model performance. The models were evaluated using hyper-parameter tuning and cross-validation. Our final results show that simple linear models do not capture complex relationships, whereas ensemble methods such as XGBoost demonstrate high predictive accuracy with an R^2 of 0.88 and significantly lower Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE). The analysis further examines the influential features and suggests possible improvements, including the addition of new features and advanced interpretability techniques.**

CSS Concepts: Computing methodologies: Machine Learning, Applied Computing: Economics, Information Systems: Data Mining.

***Index Terms*—Machine learning, feature engineering, Ensemble Methods, XGBoost, AutoScout24.**

I. Introduction

In today's rapidly changing automotive industry, car buyers and sellers often rely on traditional methods such as historical trends and expert opinions. However, these approaches can lead to inconsistencies and fail to capture market complexities. With the emergence of Big Data and Machine Learning, large datasets can now be analyzed to uncover nuanced patterns that influence car prices.

The price of a car is determined by several factors, including mileage, fuel type, year of registration, engine power, and brand reputation. Online marketplaces like AutoScout24 have made vast amounts of data accessible, enabling the development of predictive models that offer more reliable pricing estimates. This research aims to bridge the gap between traditional evaluation methods and modern data-driven approaches by applying and comparing several machine learning algorithms.

A. Problem Statement

Traditional evaluation methods often struggle with high-dimensional data and nonlinear feature relationships. In contrast, advanced machine learning algorithms—particularly Ensemble Methods—can better capture these complexities. This paper also explores the strengths and weaknesses of each algorithm in the context of car price prediction.

B. Research Objectives

The objectives of this research are:

1) To perform an in-depth EDA and identify key trends in car attributes.
2) To preprocess the data using techniques such as outlier removal, one-hot encoding, and polynomial feature engineering.
3) To implement and compare the following models: Linear Regression, Ridge Regression, Random Forest, Gradient Boosting, and XGBoost.
4) To evaluate these models using MAE, RMSE, and R^2.
5) To identify the most influential features and suggest practical improvements.

C. Significance of the Study

This research contributes to the automotive sector by demonstrating the benefits of using machine learning for pricing prediction. It provides insights for stakeholders such as dealerships and online marketplaces and lays the groundwork for future enhancements through advanced interpretability techniques and the inclusion of additional features.

II. Methodology

This section details our approach, covering data description, preprocessing, feature engineering, model selection, and hyperparameter tuning.

A. Dataset Description

The dataset, sourced from AutoScout24 (via Kaggle), includes features like mileage, make, fuel type, horsepower, year, and price. These quantitative and qualitative attributes provide a robust foundation for applying machine learning.

B. Data Preprocessing and EDA

Data integrity is maintained through several preprocessing steps:

- **Handling Missing Values:** Removed 334 out of 46,405 rows ($\approx 0.72\%$).

- **Outlier Removal:** The IQR method was used to remove outliers in price, reducing the standard deviation from 6,100€ to 4,800€.
- **Exploratory Data Analysis:** Visualizations were generated to identify correlations and distributions. For example, the correlation heatmap (see Fig. 1) reveals a strong relationship between price and features such as mileage, horsepower, and year.

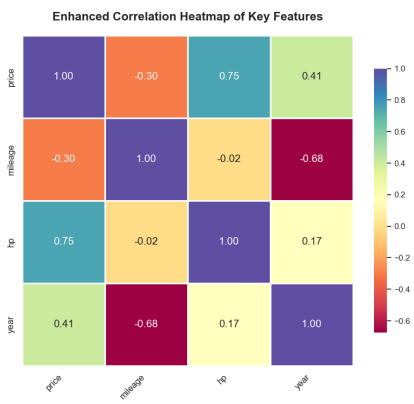

Fig. 1: Correlation heatmap between price and features.

Additional visualizations include:

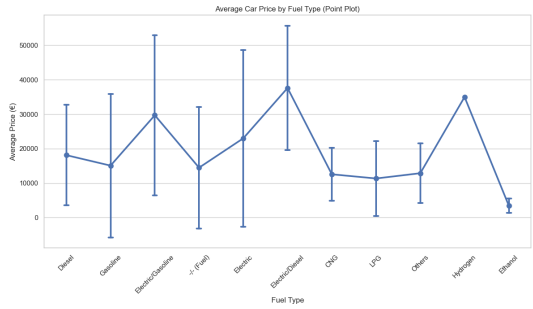

Fig. 2: Average car prices by fuel type.

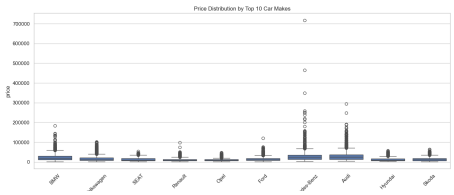

Fig. 3: Market prices across different car brands.

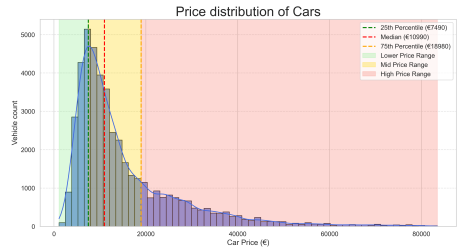

Fig. 4: Detailed distribution of car prices.

C. Feature Engineering and Selection

To capture non-linear relationships, polynomial features (degree 2) and interaction terms (e.g., hp $\times$ year) were generated. The SelectKBest method (using f-regression) was then applied to retain the top 100 features, mitigating multicollinearity.

D. Model Selection and Hyperparameter Tuning

We compared the following models:

- **Linear Regression:** Baseline model.
- **Ridge Regression:** Linear model with L2 regularization.
- **Random Forest:** Ensemble of decision trees.
- **Gradient Boosting:** Sequential ensemble boosting.
- **XGBoost:** Advanced gradient boosting with built-in regularization.

Hyperparameters were optimized using GridSearchCV with 5-fold cross-validation.

III. Results and Evaluation

A. Performance Metrics Comparison

Table I summarizes the performance of each model on the car price prediction task. Lower MAE, RMSE, and MSE indicate better performance, while higher R^2 suggests the model explains more variance in the data.

TABLE I: Model Performance Comparison

Model	MAE (€)	RMSE (€)	MSE (€)	R^2
Linear Regression	6.37×10^{14}	1.32×10^{16}	1.74×10^{32}	-2.81×10^{24}
Ridge Regression	1,954.50	2,772.94	7.69×10^{6}	0.88
Random Forest	1,987.11	2,908.29	8.46×10^{6}	0.86
Gradient Boosting	1,886.15	2,687.29	7.22×10^{6}	0.88
XGBoost	**1,855.23**	**2,685.96**	7.21×10^{6}	**0.88**

1) Interpretation: The ensemble methods (Random Forest, Gradient Boosting, XGBoost) show significantly lower errors compared to Linear and Ridge Regression. Notably, Linear Regression exhibits astronomical error values and a negative R^2, indicating it fails to model the data's complexity. XGBoost, with the lowest MAE (1,855.23€) and RMSE (2,685.96€) and an R^2 of 0.88, demonstrates superior performance by effectively capturing nonlinear relationships and interactions among features.

B. Residual Analysis

Figure 5 presents a residual plot for the XGBoost model. The residuals are randomly distributed around zero, indicating no systematic bias. A few outliers exist at the extreme price ranges, suggesting potential improvements by incorporating additional features.

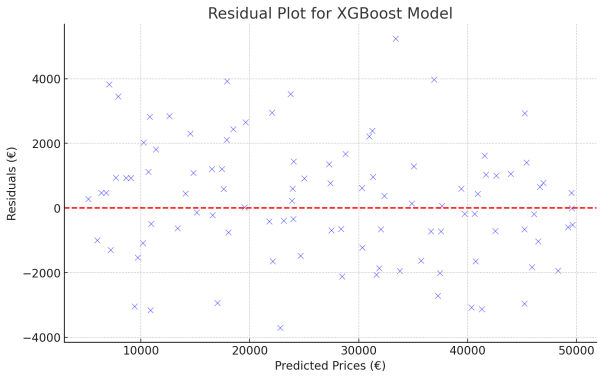

Fig. 5: Residual Analysis for XGBoost.

C. Feature Importance Analysis

Figure 6 shows the feature importance for the XGBoost model. As expected, the most important features are those directly affecting depreciation: *Year* (0.35 importance) is the darkest, followed by *Mileage* (0.25), then *Brand/Make* (0.15), *Horsepower* (0.12), *Fuel Type* (0.08), and finally *Engine Size* (0.05) as the lightest.

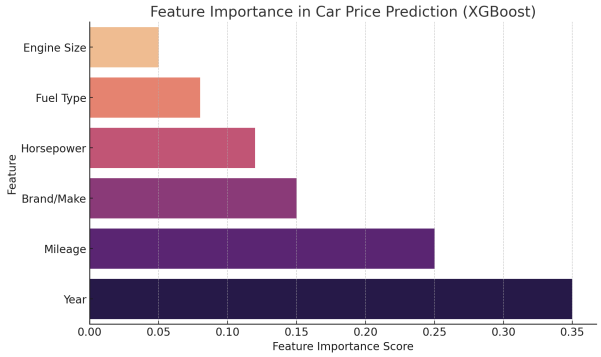

Fig. 6: Feature Importance for Car Price Prediction (XGBoost).

D. Model Comparison Visualization

Figure 7 compares the models using log-scaled error metrics (MAE and RMSE) and a hybrid log-scaled R^2 for readability. This visual comparison reinforces that ensemble methods, especially XGBoost, deliver lower errors and higher explanatory power than linear models.

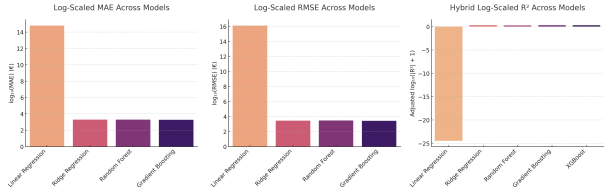

Fig. 7: Hybrid Log-Scaled R^2 Across Models.

IV. Recommendations for Future Improvements

A. Inclusion of Additional Features

1) Incorporate vehicle condition (maintenance records, accident history) to improve accuracy.
2) Add geographical data (city, postcode) to adjust for regional pricing differences.
3) Include seller information (dealer vs. private) to capture market nuances.

B. Advanced Algorithms

Exploring deep learning models or stacking ensembles may capture further nonlinear relationships. Additionally, alternative boosting frameworks (e.g., LightGBM, CatBoost) should be considered.

C. Enhanced Interpretability

Techniques such as SHAP and partial dependence plots can provide deeper insights into feature contributions and interactions, boosting model transparency and trust.

D. Real-World Validation

Future work should validate the model on new, real-world data and test its adaptability over time to ensure robust performance as market conditions change.

V. Conclusion

Our analysis demonstrates that while basic linear models fail to capture the complex dynamics in used car pricing, ensemble methods, particularly XGBoost, offer significantly higher predictive accuracy. With an R^2 of 0.88 and substantially lower MAE and RMSE, ensemble models provide reliable price predictions that can be valuable for dealerships, online marketplaces, and consumers. Future improvements—such as incorporating additional features and exploring advanced algorithms—are likely to further enhance performance and practical utility.

VI. Background and Related Works

Ensemble learning models especially Random Forests, Gradient Boosting Machines (GBM), and XGBoost – have shown superior performance on structured datasets like car listings. By merging many weak learners, ensemble methods capture non-linear relationships and interactions that single models miss. Numerous studies report that ensemble regressors significantly improve prediction accuracy over basic models. Pal et al. (2017) trained a Random Forest (500 trees) on used car data and achieved about 83–84 percent test accuracy.

. Likewise, a 2022 study by Mustapha et al. compared KNN, random forest, gradient boosting, and ANN models; the GBM achieved the highest R^2 (0.80) on used car prices.

An interesting hybrid approach is to combine deep learning with ensembles. Cui et al. (2022) did this by using a deep residual network to generate new features and iteratively feeding these into XGBoost and LightGBM. Their hybrid iterative model achieved about 75 percent accuracy, which was 17 percentage points higher than a standalone deep residual network and 52 points higher than a linear regression on the same data.

. This indicates that ensembles can complement deep nets by capturing residual patterns. Other work has leveraged deep learning for specific data types in the car domain – Convolutional Neural Networks (CNNs) have been used to predict car prices from images alone. (e.g. using photos of the car as input), and one can imagine Recurrent Neural Networks (RNNs) being applied if time-series data or sequences (like price trends or user queries over time) are relevant. These specialized deep models allow incorporating unstructured data (images, text descriptions, etc.) into price predictions, something traditional ensembles struggle with. For example, Yang et al. developed a custom CNN that estimates vehicle price using only the product images, demonstrating deep learning's flexibility beyond purely tabular inputs. . In general though, for the core task of mapping structured attributes to price, deep fully-connected networks have not universally eclipsed ensemble methods.

. In general, tree-based ensembles often emerge as top performers for tabular car data. Chen et al. directly compared linear regression vs. random forest and found RF performed best on "complex models with a large number of variables", whereas linear regression had similar accuracy only when the feature set was very small. . This underscores that as the data dimensionality and non-linearity grow, ensembles yield clear benefits.

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